Ashwin Padaki

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Summary

Computer Science Ph.D. student specializing in efficient data structures for nearest neighbor search and other problems involving high dimensional data. Also interested in practical aspects of vector retrieval systems.

Education

University of Pennsylvania — Philadelphia, PA
Ph.D in Computer and Information Science — *2024 – 2029 (expected)*

Advisors: Erik Waingarten and Sanjeev Khanna.

Columbia University — New York, NY
B.A. in Computer Science, Mathematics — *2020 – 2024*

Industry Experience

Research Scientist Intern — Jun 2026 – present
Pinecone — *New York, NY*

- Building an open-source vector quantization benchmark and developing new algorithms.

Quantitative Trader Intern — Jun 2022 – Aug 2022
Optiver — *Chicago, IL*

Publications

Unless otherwise specified (∗), authors are ordered alphabetically by last name.

1. **Ashwin Padaki**, Amir Ingber, Edo Liberty. VQ-bench: A Composable Vector Quantization Framework.
   VecDB@VLDB 2026

2. Sanjeev Khanna, **Ashwin Padaki**, Erik Waingarten. Learning Partition Trees for Nearest Neighbor Search.
   (in submission)

3. (∗) Tian Zhang, **Ashwin Padaki**, Jiaming Liang, Zack Ives, Erik Waingarten. Prune, Don't Rebuild: Efficiently Tuning α-Reachable Graphs for Nearest Neighbor Search.
   (in submission)

4. Sanjeev Khanna, **Ashwin Padaki**, Erik Waingarten. Sparse Navigable Graphs for Nearest Neighbor Search: Algorithms and Hardness.
   SODA 2026

5. Sanjeev Khanna, **Ashwin Padaki**, Krish Singal, Erik Waingarten. A Polynomial Space Lower Bound for Diameter Estimation in Dynamic Streams.
   FOCS 2025

6. Henry Fleischmann, Kyrylo Karlov, Karthik C. S., **Ashwin Padaki**, Styopa Zharkov. Inapproximability of Maximum Diameter Clustering for Few Clusters.
   SODA 2025

7. Josh Alman, Yunfeng Guan, **Ashwin Padaki**. Smaller Low-Depth Circuits for Kronecker Powers.
   SODA 2023

Talks

1. **Learning Partition Trees for Nearest Neighbor Search**
   - NYU Algorithms Seminar — May 2026
2. **Sparse Navigable Graphs for Nearest Neighbor Search**
   - SODA 2026 Conference — Jan 2026
   - NYC Graduate Student Theory Seminar — Nov 2025
   - DIMACS/Rutgers Theory Seminar — Nov 2025
   - CMU Theory Seminar — Sep 2025
3. **Inapproximability of Maximum Diameter Clustering for Few Clusters**
   - SODA 2025 Conference — Jan 2025

Service

Head Teaching Assistant — Sep 2025 – May 2026
Algorithms for Big Data (UPenn)

Seminar Organizer — Sep 2025 – present
UPenn Theory Seminar

Seminar Instructor — Sep 2023 – Dec 2023
Columbia Undergraduate Learning Seminar in Theoretical Computer Science

Achievements

National Science Foundation (NSF) Graduate Research Fellow — 2024
Phi Beta Kappa Inductee — 2024
Putnam Mathematical Competition, Top 500 Scorer — 2022